



Lesson 8: Final Project — Our Own Mini Game!



Welcome to Your Final Mission!

You've learned SO much: variables, input, if/else decisions, loops, and lists. Now it's time to use ALL of it to build your very own original mini game!



Think Like a Coder

Great game designers PLAN before they code. Before typing anything, think: what does the player do? What are the rules? How do they win or lose?



Step 1: A Model Game — The Lucky Number Game

Let's look at an example game together:

```
import random

secret_number = random.randint(1, 5)
guess = int(input("Guess a number between 1 and 5: "))

if guess == secret_number:
    print("🎉 You guessed it! You win!")
else:
    print(f"Not quite! The number was {secret_number}. Try again next time!")
```

Run it a few times. Sometimes you'll win, sometimes you won't — that's the randomness at work!



Robot Tip

`random.randint(1, 5)` picks a random WHOLE number between 1 and 5, including both 1 and 5. Remember to `import random` at the very top of your file!

Mini Challenge

Add a replay loop so the player can guess again and again:

```
import random

play_again = "yes"
while play_again == "yes":
    secret_number = random.randint(1, 5)
    guess = int(input("Guess a number between 1 and 5: "))
    if guess == secret_number:
        print("🎉 You guessed it! You win!")
    else:
        print(f"Not quite! The number was {secret_number}.")
    play_again = input("Play again? (yes/no) ")

print("Thanks for playing!")
```

Make It Yours: Design Your Own Game!

Use the Game Planning Sheet in [Activities.md](#) to design your OWN original game. Here are your building blocks:

- **Variables** — to store information
- **input()** — to ask the player questions
- **if/elif/else** — to make decisions
- **while / for loops** — to repeat things
- **lists + random** — to add variety and surprises

Some ideas to inspire you: - A rock-paper-scissors game against the computer - A "would you rather" question game - A treasure hunt where the player guesses a hidden location - Your very own version of the Word Guessing Game from Project 06!

Bonus Challenge

Add a scoring system or limited number of attempts to your game, like this:

```
import random

secret_number = random.randint(1, 5)
attempts = 3
guess_count = 0
won = False

while guess_count < attempts and not won:
    guess = int(input(f"Guess a number 1-5 (attempt {guess_count + 1} of {attempts}): "))
    guess_count = guess_count + 1
    if guess == secret_number:
        print("🎉 You guessed it! You win!")
        won = True
    else:
        print("Not quite, try again!")

if not won:
    print(f"Out of attempts! The number was {secret_number}.")
```



Debug Detective

Even the best coders make mistakes! When you test your own game: 1. Run it at least 3 times. 2. Try answers that should win AND answers that should lose. 3. If you see a red error message, read it carefully — it usually tells you exactly which line has the problem. 4. Fix one thing at a time, then run it again.



Victory Moment

You just built your very own game — completely from your own ideas — using everything you learned in this whole course. You are officially a Python game builder!



What's Next?

You've completed all 8 lessons! Now it's time to build your 3 big portfolio projects: a Personal Quiz App, a Simple Calculator, and a Word Guessing Game. Check them out in the [Projects](#) folder — you're ready for all of them!